

Bangladesh Army University of Science and Technology (BAUST), Saidpur.



Lab Report

Department: Computer Science and Engineering.

Course Title: Electrical Drives and Instrumentation Sessional.

Course Code: EEE 2270

Experiment No: 04

Name of the Experiment: Open Circuit Test and Short Circuit Test of a transformer.

Submitted By

Submitted To

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Date of Submission: 26/03/2024

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Signature:

Experiment No – 3

Experiment Name: Open Circuit Test and Short Circuit Test of a transformer.

Objective:

1. To determine core loss of a transformer using Open Circuit Test.
2. To determine copper loss of a transformer using Short Circuit Test.

Apparatus:

1. Transformer.
2. Wattmeter.
3. Voltmeter.
4. Ammeter.
5. Universal power supply.
6. Connecting wire.

Theory:

Open Circuit Test: The open circuit test, also known as the no-load test, is a method used to assess the core losses in a transformer. This test is performed by disconnecting the transformer from any load and applying a variable AC voltage to its primary winding while keeping the secondary winding open. The purpose of the test is to determine the core losses, which primarily consist of hysteresis and eddy current losses occurring in the transformer's magnetic core.

Working principle of Open Circuit Test:

This test is performed on the primary winding of the transformer. The wattmeter, ammeter, and voltmeter are connected to their primary winding. The nominal rated voltage is supplied to their primary winding with the help of the ac source. The secondary winding of the transformer is kept open, and the voltmeter is connected to their terminal. This voltmeter measures the secondary induced voltage. As the secondary of the transformer is open, thus no-load current flows through the primary winding. The value of no-load current is very small as compared to the full rated current. The copper loss occurs only on the primary winding of the transformer because the secondary winding is open. The reading of the wattmeter only represents the core and iron losses. The core losses of the transformer are the same for all types of loads.

Short Circuit Test: The short-circuit test is a procedure conducted on a transformer to determine its impedance, losses, and other parameters when subjected to a short-circuit condition. This test is crucial for assessing the transformer's performance and ensuring it meets design specifications.

Working principle of Short Circuit Test:

Short circuit test in the transformer is conducted on the high voltage side and low voltage side is short circuited. In this test, we need to increase the applied voltage gradually towards rated voltage up to rated load current passing in the secondary short- circuited winding by keeping frequency constant. In HV winding, rated current is less than LV winding. As rated current is less on HV side, it is convenient to conduct this test on HV side by short circuiting the LV terminals. This ensures that the low range of meters can be used for conducting this test as

wattmeter is connected to the primary side (High voltage side). Therefore, the primary side is generally chosen as primary side. Short circuit test can be performed on either side but generally performed on HV side.

Circuit Diagram:

Data Table:**For Open Circuit Test:**

Input Voltage, V_{oc}	110
Input Current, I_{oc}	0.17
Input Power, P_{oc}	6
Power Factor = $\frac{P_{oc}}{V_{oc} \cdot I_{oc}}$	0.32

Discussion: Both the short-circuit test and open-circuit test were deemed essential for characterizing the performance of the transformer. During the short-circuit test, a low voltage was applied to the secondary winding, resulting in a short circuit, inducing a high current flow through the windings. This facilitated the measurement of impedance and copper losses, providing valuable insights into the transformer's ability to withstand short-circuit stresses and its thermal behavior under fault conditions. Conversely, in the open-circuit test, a low voltage was applied to the primary winding while the secondary winding remained open. Measurements were taken to determine core losses and magnetizing current. This test served to assess the transformer's core losses and magnetizing characteristics under no-load conditions. The data obtained from both tests were analyzed to optimize transformer design, ensure compliance with efficiency standards, and enhance overall performance. By understanding the transformer's behavior under different load conditions, engineers were able to design more reliable and efficient power distribution systems. The results of these tests contributed significantly to the assessment and improvement of transformer performance in diverse energy applications.

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